

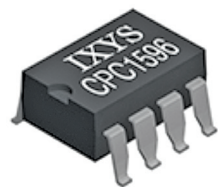
Devices replaces discrete implementations by integrating an optical PPG subsystem to measure electrocardiogram (ECG or EKG), heart rate (ECG or optical PPG), blood-oxygen saturation (SpO₂), and respiration rate (using BioZ). It permits small vital sign devices by fitting those multiple functions into a 2.6mm x 2.8mm package.

Maxim Integrated

www.maximintegrated.com

Load-biased gate driver

Littelfuse has launched a high-voltage optically-isolated MOSFET gate driver that requires no external power supply, yet can provide fast load turn-on speeds



in the order of tens of microseconds. The CPC1596 570V load-biased gate driver is ideal for use in a wide

range of power electronics applications including industrial and medical devices. It provides fast turn-on switch speed (typically 40µs), which enables designers to create a custom, higher power, optically isolated, DC-only, or AC-DC solid-state relay (SSR) with external MOSFET(s). The use of load-biasing eliminates the need for an opto-gate driver auxiliary power supply.

Littelfuse

www.littelfuse.com

Laser diode for lidar

ROHM has developed a high optical output laser diode, the RLD90QZW3, for applications such as automated guided vehicles (AGVs) and service robots in the industrial equipment sector and robot vacuums in the consumer field that incorporate lidar for distance measurement and spatial recognition. A beam having width of 225µm is emitted,



which is 22% narrower than conventional beams. At the same time, uniform emission intensity, together with low temperature dependence of the laser wavelength,

ensure stable performance, contributing to higher accuracy and longer distance reach in various lidar applications.

ROHM Semiconductor

www.rohm.com

MEMS barometric pressure sensor

TDK has announced worldwide availability of the InvenSense ICP-20100 platform, a new-generation, feature-rich barometric pressure sensor that



is ideal for applications in smartphones, tablets, drones, and smart home appliances.

The device combines a barometric pressure and a temperature sensor in a small 2.0mm x 2.0mm x 0.8mm package and enables flexible user designs with multiple input voltage levels including 1.2V, 1.8V, and 3.3V, and a choice of I²C, I³CSM, and SPI interfaces. It can be configured to achieve ultra-low noise or ultra-low power performance.

TDK Corporation

www.tdk.com

Flyback switcher ICs

Power Integrations has launched the InnoSwitch3-PD family of ICs for USB Type-C, USB power delivery (PD), and USB programmable power supply (PPS) adaptors. This compact InSOP-24D-packaged IC includes a USB-C and PD controller, a high-voltage PowiGaN switch, a multi-mode quasi-resonant

flyback controller, secondary-side sensing, FluxLink isolated digital feedback, and a synchronous-rectification



driver. Featuring no-load power consumption as low as 14mW, power supply designs using InnoSwitch3-

PD ICs meet all global energy-efficiency regulations. The high efficiency of these devices ensures low heat dissipation, eliminating the need for bulky heatsinks.

Power Integrations

www.power.com

Sub-GHz wireless SoCs

Silicon Labs has launched new sub-1GHz (sub-GHz) SoCs, the EFR32FG23 (FG23) and EFR32ZG23 (ZG23), thus delivering sub-GHz wireless solutions



that combine long-range RF and energy efficiency with certified Arm PSA Level 3 security to meet the global demand for

high-performance, battery-powered IoT products. The SoC solutions provide developers with flexible, multiprotocol sub-GHz connectivity options supporting a wide range of modulation schemes and advanced wireless technologies, including Amazon Sidewalk, mioty, Wireless M-Bus, Z-Wave, and proprietary IoT networks.

Silicon Labs

www.silabs.com

MODULES & BOARDS

Digital power supply boards

The STEVAL-NRG011TV board from STMicroelectronics helps designers quickly build 200W digital power supplies and adaptors for LED and OLED televisions to exceed the toughest ener-



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